

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|--|-------|---|
| A) Explain the operations involved in the planning of new project. | (CO2) | 6 |
| B) Differentiate between the wheel and track mounted tractor. | (CO1) | 6 |
| C) What is bulldozer? State the operation carried out by bulldozer and factors affecting the selection of bulldozer. | (CO1) | 6 |

Q.2 Solve Any Two of the following.

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|---|-------|---|
| A) Define explosives and explain the types of explosives. | (CO2) | 6 |
| B) Explain various drills based on the application of mechanical energy used in excavation. | (CO2) | 6 |
| C) Discuss the necessity of drainage in excavation. | (CO2) | 6 |

Q. 3 Solve Any Two of the following.

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|--|-------|---|
| A) Explain the operations involved in RMC plant. | (CO2) | 6 |
| B) Illustrate the types of concrete mixer. | (CO1) | 6 |
| C) Write the short note on grouting. | (CO2) | 6 |

Q.4 Solve Any Two of the following.

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|--|-------|---|
| A) State the advantages and disadvantages of prefabricated construction. | (CO2) | 6 |
| B) Illustrate the types of tower cranes. | (CO1) | 6 |
| C) Explain the operations involved in hot mix plan with layout. | (CO1) | 6 |

Q. 5 Solve Any Two of the following.

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|--|-------|---|
| A) Explain factors affecting the site selection for bridge construction. | (CO1) | 6 |
| B) Illustrate diaphragm wall. | (CO2) | 6 |
| C) Describe safety measures in construction. | (CO2) | 6 |